

Multivariable calculus

1. Vectors, lines, planes

1.0.1. $\mathbb{R}^n$

Definition 1.0.1 (Two-dimensional real-coordinate space ($\mathbb{R}^2$))

$$\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}$$

Definition 1.0.2 (Three-dimensional real-coordinate space ($\mathbb{R}^3$))

$$\mathbb{R}^3 = \{(x, y, z) : x, y, z \in \mathbb{R}\}$$

x, y, z should be presented such that the coordinate system is right-handed ($\hat{k} = \hat{i} \times \hat{j}$ should have direction according to the right-hand rule).

Definition 1.0.3 (Standard basis vectors)

The standard basis vectors of a space are the unit vectors that go along the axes of the space. All vectors in that space can be expressed as sums of scalar multiples of the standard basis vectors of that space.

The standard basis vectors of $\mathbb{R}^2$ are $\hat{i}$ and $\hat{j}$, also called e_1 and e_2 .

The standard basis vectors of $\mathbb{R}^3$ are $\hat{i}$, $\hat{j}$, and $\hat{k}$, also called e_1 , e_2 , and e_3 .

1.1. Vector space

Definition 1.1.1 (Vector space)

A vector space over a field F is a set V with the following operations:

- **Vector addition:** $u \in V, v \in V \mapsto (u + v) \in V$, which satisfies the properties of addition

- **Scalar multiplication:** $\alpha \in F, v \in V \mapsto \alpha v \in V$, which satisfies the properties of scalar multiplication

Definition 1.1.2 (Properties of addition)

For all $u, v, w \in V$:

- (A1) Commutativity: $u + v = v + u$
- (A2) Associativity: $(u + v) + w = u + (v + w)$
- (A3) Identity: $\exists 0 \in \mathbb{R}$ s.t. $0 + u = u$
- (A4) Additive inverse: For $u \in F$, $\exists -u \in F$ s.t. $u + (-u) = 0$

Definition 1.1.3 (Properties of scalar multiplication)

For all $\alpha, \beta \in F, v, w \in V$:

- (S1) Associativity: $(\alpha\beta)v = \alpha(\beta v)$
- (S2) Distributivity over scalar addition: $(\alpha + \beta)v = \alpha v + \beta v$
- (S3) Distributivity over vector addition: $\alpha(v + w) = \alpha v + \alpha w$
- (S4) Multiplicative identity: $1v = v$

Theorem 1.1.4

$\mathbb{R}^n$ is a vector space over the field $\mathbb{R}^n$.